

DriveEase:Functional Requirements Document (FRD)

1. Introduction

1.1 Purpose

The purpose of this application is to provide an online platform where users can search, compare, and book cars on an **hourly or daily basis**.

Users will be able to select a vehicle based on location, date, duration, vehicle type, and price. The system will provide vehicle availability, pricing, online payment, booking confirmation, and booking management.

An administrative panel will allow administrators to manage users, vehicles, bookings, pricing, payments, and promotional offers. This allow user to travel according to their wish and saves heavy taxi prices.

1.2 Scope

The application will provide:

- User registration and login
- Car search and filtering
- Hourly car booking
- Daily car booking
- Vehicle availability management
- Dynamic pricing
- Online payment
- Booking confirmation
- Booking history
- Booking modification
- Notifications
- Ratings and reviews
- Admin dashboard
- Vehicle management
- User management

- Booking management
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2. User Roles

2.1 Customer

The customer can:

- Register and log in.
- Search for available cars.
- View car details.
- Select hourly or daily rental.
- Make a booking.
- Make online payments.
- View booking details.
- View booking history.
- Manage their profile.
- Rate and review vehicles.

2.2 Administrator

The administrator can:

- Manage customers.
- Manage vehicles.
- Manage vehicle availability.
- Manage bookings.
- Manage pricing.
- Manage payments.
- Manage promotional offers.
- Manage reviews.
- View dashboard information.

2.3 Car Operator/Owner

If third-party vehicle owners are supported, the operator can:

- Register vehicles.
- Add vehicle details.

- Upload vehicle documents.
- Set vehicle availability.
- View bookings.
- View earnings.
- Update vehicle status.

3. Functional Requirements

FR-01: User Registration

The system shall allow a new customer to create an account.

Requirements

- Customer shall provide:
 - Full name
 - Mobile number
 - Email address
 - Password
- The system shall validate the provided information.
- The system shall verify the user's mobile number/email where applicable.
- The system shall prevent duplicate accounts.
- The system shall display an error message for invalid information.

FR-02: User Login

The system shall allow registered users to log in.

Requirements

- User shall enter email/mobile number and password.
- The system shall validate the credentials.
- Successful users shall be redirected to the home/dashboard page.
- The system shall provide a "Forgot Password" option.
- The system shall securely manage user sessions.

FR-03: Car Search

The system shall allow users to search for available cars.

Search criteria

Users shall be able to specify:

- Pickup location
- Pickup date
- Pickup time
- Rental duration
- Booking type:
 - Hourly
 - Daily

The system shall display vehicles available according to the selected criteria.

FR-04: Car Listing

The system shall display available vehicles in a searchable list.

Each vehicle listing shall contain:

- Vehicle image
- Vehicle name/model
- Vehicle type
- Seating capacity
- Fuel type
- Transmission type
- Rental price
- Availability
- Pickup location
- Basic terms and conditions

Users shall be able to sort vehicles by:

- Price
 - Vehicle type
 - Rating
 - Popularity
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FR-05: Car Filtering

The system shall allow users to filter available cars.

Filters may include:

- Price range
- Car type
- Number of seats
- Fuel type
- Transmission
- Brand
- Rating
- Pickup location

The system shall update the vehicle list according to the selected filters.

FR-06: Car Details

The system shall provide a detailed page for each vehicle.

The page shall display:

- Multiple vehicle images
- Vehicle model
- Vehicle type
- Seating capacity
- Fuel type
- Transmission
- Features
- Rental price
- Security deposit, if applicable
- Pickup/return information
- Cancellation policy
- Customer ratings and reviews

The user shall be able to proceed to booking from the vehicle details page.

4. Booking Management

FR-07: Hourly Booking

The system shall allow customers to rent a vehicle for a selected number of hours.

Example

Car: Sedan

Pickup: 10:00 AM

Duration: 5 hours

The system shall calculate the total rental amount based on the applicable hourly rate and additional charges.

FR-08: Daily Booking

The system shall allow customers to rent a vehicle for one or more days.

Example

Pickup: 12 August, 10:00 AM

Return: 15 August, 10:00 AM

The system shall calculate the rental cost according to the applicable daily rate.

FR-09: Vehicle Availability

Before confirming a booking, the system shall verify vehicle availability.

The system shall prevent:

- Double booking
- Booking an unavailable vehicle
- Booking a vehicle during maintenance
- Booking outside the vehicle's available operating period

Once a booking is confirmed, the selected vehicle shall be marked as unavailable for the corresponding period.

FR-10: Booking Summary

Before payment, the system shall display a booking summary containing:

- Customer information
- Vehicle information
- Pickup location
- Pickup date/time
- Return date/time
- Rental duration
- Base rental price
- Taxes
- Additional charges
- Discount/promo code
- Security deposit, if applicable
- Final payable amount

The customer shall confirm the details before proceeding to payment.

FR-11: Dynamic Pricing

The system shall calculate rental prices based on configurable pricing rules.

Pricing may depend on:

- Hourly/day rate
- Rental duration
- Vehicle type
- Demand
- Peak/off-peak period
- Weekend/holiday
- Pickup location
- Promotional discount

The administrator shall be able to configure pricing rules.

FR-12: Promo Codes and Discounts

The system shall allow customers to apply promotional codes during checkout.

The system shall:

- Validate the promo code.
- Check expiry date.
- Check usage limits.
- Check minimum booking amount.
- Calculate the applicable discount.
- Display the updated total.

Invalid or expired promo codes shall generate an appropriate message.

5. Payment Management

FR-13: Online Payment

The system shall allow customers to pay for their bookings online.

Supported payment methods may include:

- Credit/Debit Card
- UPI
- Net Banking
- Digital Wallets

The system shall generate a unique transaction ID for each payment.

FR-14: Payment Confirmation

After successful payment:

- The booking shall be confirmed.
- A unique booking ID shall be generated.
- Payment status shall be updated.
- The customer shall receive a booking confirmation.
- The booking shall appear in the customer's booking history.

If payment fails, the booking shall not be confirmed.

6. Booking Management

FR-15: Booking History

The customer shall be able to view their previous and current bookings.

The booking history shall contain:

- Booking ID
- Vehicle details
- Pickup location
- Pickup date/time
- Return date/time
- Rental duration
- Booking amount
- Payment status
- Booking status

The customer shall be able to view individual booking details.

7. Customer Dashboard

FR-16: My Bookings

The customer shall be able to view:

- Upcoming bookings
- Active bookings
- Completed bookings
- Cancelled bookings

Each booking shall display:

- Booking ID
- Vehicle
- Pickup/return details
- Duration
- Amount
- Payment status

- Booking status

FR-17: Booking Details

The customer shall be able to open an individual booking and view complete information.

The system shall provide options such as:

- View booking
- Download invoice
- Contact support

8. Unique Functional Requirement

FR-18: Smart Car Recommendation

The system shall provide personalized vehicle recommendations based on the customer's requirements and previous booking behavior.

The recommendation system may consider:

- Previous vehicle selections
- Rental duration
- Number of passengers
- Budget
- Preferred vehicle type
- Pickup location
- Frequently selected features
- Vehicle rating

Example

If a customer selects:

4 passengers + 3-day booking + ₹2,500 budget

the system can recommend vehicles that provide the best combination of:

Price + passenger capacity + rating + availability.

The system shall display a **"Recommended for You"** section in the search results.

9. Notifications

FR-19: Booking Notifications

The system shall send notifications for important booking events.

Notifications may include:

- Registration confirmation
- Booking confirmation
- Payment confirmation
- Booking reminder
- Booking status update
- Refund/payment-related notification

Notifications may be delivered through:

- Email
 - SMS
 - Push notification
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10. Ratings and Reviews

FR-20: Vehicle Rating and Review

After completing a booking, the customer shall be able to rate the vehicle/service.

The customer can provide:

- Star rating
- Written review

The system shall prevent users from reviewing vehicles they have not booked.

The administrator shall be able to moderate reviews.

11. Admin Functional Requirements

FR-21: Admin Dashboard

The administrator shall have access to a dashboard displaying:

- Total users
 - Total vehicles
 - Active bookings
 - Completed bookings
 - Cancelled bookings
 - Revenue
 - Pending payments
 - Vehicle availability
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FR-22: Vehicle Management

The administrator shall be able to:

- Add vehicles.
 - Edit vehicle information.
 - Delete/deactivate vehicles.
 - Upload vehicle images.
 - Set rental prices.
 - Set vehicle availability.
 - Mark vehicles as under maintenance.
 - View vehicle booking history.
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FR-23: User Management

The administrator shall be able to:

- View users.
- Search users.
- View user booking history.

- Activate/deactivate accounts.
 - Manage user verification.
 - View customer activity.
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FR-24: Booking Management

The administrator shall be able to:

- View all bookings.
 - Search bookings.
 - Filter bookings.
 - Modify booking status.
 - Cancel bookings.
 - Manage payment status.
 - View complete booking details.
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12. Non-Functional Requirements

12.1 Performance

- Search results should load quickly.
- The system should support multiple concurrent users.
- Vehicle availability should be updated in near real time.

12.2 Security

- Passwords shall be securely stored.
- Payment information shall be handled through a secure payment gateway.
- Role-based access control shall be implemented.
- Customer information shall be protected.
- The system shall prevent unauthorized access.

12.3 Availability

The application should be available 24/7 except during scheduled maintenance.

12.4 Scalability

The system should support future expansion to:

- Multiple cities
- More vehicles
- Multiple vehicle operators
- Mobile applications
- Corporate customers